Md Shahanur Islam Shagor
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I am writing to apply for the Embedded Software-Entwicklung position at REVOTION GmbH. The idea of building firmware that directly powers smart systems in caravans, boats, and yachts is exactly the kind of work I want to be part of. I appreciate that your team values real ownership over features, from the first commit to a product in the field.

My technical foundation is in C++ and Python, with practical experience developing embedded systems using ESP32 and Raspberry Pi. I have worked with real-time data streaming, sensor integration, and hardware bring-up. In my project "Wireless Vision-Aid for the Blind," I built a complete embedded system that combined an ESP32-CAM for video capture with a Raspberry Pi running YOLOv8n for object detection. I implemented a custom UDP streaming protocol to achieve low-latency communication between the camera module and the processing unit, and optimized the entire pipeline to run offline on constrained hardware.

This role involves developing firmware for BRAIN and NODE modules using C on ESP32-S3 with FreeRTOS/ESP-IDF, designing BLE interfaces, and integrating third-party devices over CAN, RS-485, and Modbus. While my direct experience is with C++ and Python rather than C, I am confident in my ability to transition quickly given my strong systems programming background and hands-on work with microcontrollers. I am also familiar with Git, structured code reviews, and debugging with logic analyzers and oscilloscopes.

What stands out to me about REVOTION is the combination of embedded systems work with real-world impact. The chance to see your code running on vehicles in active use, and to shape the firmware architecture in a small, focused team, is rare. I also bring experience with AI at the edge, which aligns with your nice-to-have for Edge-AI and TinyML on microcontrollers.

In addition to my technical experience, I work effectively across multidisciplinary teams and communicate clearly with both technical and non-technical stakeholders. I am fluent in Bengali and English, have working knowledge of Russian and Hindi, and enjoy collaborating in international and cross-cultural environments. I have attached my CV for your review and would welcome the opportunity to discuss how I can contribute to your team.

Sincerely,

Md Shahanur Islam Shagor
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